

A Visual Based Algorithm for Measuring the Speed of Scrap Metal Vibration Feeding

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Abstract

In the field of steel metallurgy, scrap metal logistics in electric arc furnaces are typically conveyed horizontally through inertia-based vibratory conveyors. However, due to the different types, sizes, and appearances of scrap metal, as well as the repeated horizontal vibrations of conveyors, it is difficult to identify specific types of scrap metal for speed measurement. This article proposed a visual based speed measurement algorithm for the vibration feeding scenario of scrap metal. Firstly, we preprocessed the video dataset during the vibration feeding of scrap metal. Then we constructed a feature point extraction and matching model. Finally, based on the length of the feeding area, we estimated the vibration feeding speed by matching the pixel displacement of feature points in two frames in unit time. After on-site application verification, the speed curve measured by the algorithm is reasonable and consistent with the frequency curve of the horizontal vibrator, solving the problem of measurement of vibration feeding speed.

CCS Concepts

• Computing methodologies; • Artificial intelligence; • Computer vision; • Computer vision problems;

Keywords

Image processing, Scrap metal logistics, Feature point extraction and matching

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1 Introduction

In recent years, with the popularization and application of deep learning, speed measurement algorithms based on this technology have been widely applied in fields such as vehicle speed measurement and conveyor speed measurement. In most scenarios, a detect-then-track [1–3] approach is commonly employed. However, in the process of scrap metal logistics, it is difficult to identify and track specific types of scrap metal due to the differences in types, sizes, and appearances of scrap metal in the feeding area, as well as issues such as stacking and obstruction, making it difficult to achieve the goal of speed measurement. Additionally, the use of inertia-based vibratory conveyors for horizontal vibration transmission during transportation further complicates the measurement.

In addition to object detection and tracking methods, the extraction and matching of local features [4–6] such as corner points enables the calculation of pixel displacement without considering the type of object, which is more suitable for the scenario of scrap metal vibration feeding. In deep learning based feature point extraction and matching methods, Key.Net [7] combines handcrafted and learned CNN filters within a shallow multi-scale architecture and proposes a efficient trainable detector, MagicPoint [8] proposed a method for feature point detection on synthetic datasets. In addition, many methods integrate feature detection into the entire matching pipeline by jointly training with feature description and matching [9–11], which can enhance the final matching performance and optimize the entire procedure.

In view of the situation and challenges above, we focus on using deep learning based feature point extraction and matching algorithms to solve the speed measurement problem, and propose a visual based algorithm for measuring the speed of scrap metal vibration feeding. By preprocessing the video stream of scrap metal feeding data and utilizing the SuperPoint [12] feature point extraction model along with the SuperGlue [13] feature point matching model, the scrap metal feeding speed can be estimated based on the length of the feeding area and the pixel displacement in unit time.

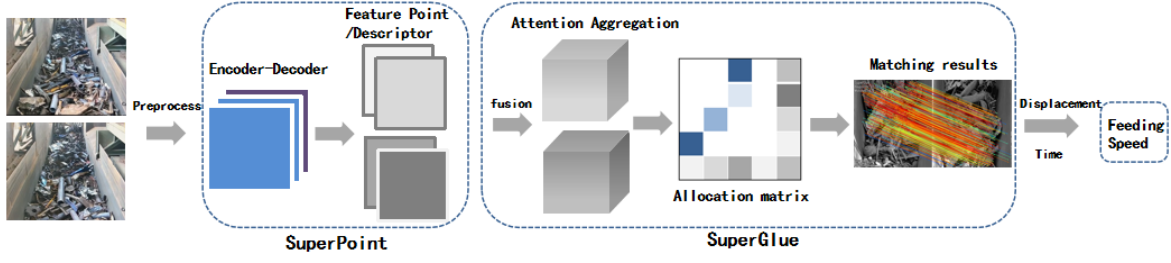


Figure 1: The overall pipeline of algorithm for measuring the vibration feeding speed of scrap metal

2 Related Works

2.1 Feature Points Detection and Description with SuperPoint

SuperPoint [12] is a self-supervised feature points detection and descriptor extraction model. The model renders a large-scale synthetic graphics dataset composed of basic shapes such as triangles, rectangles, and cubes, which can be easily annotated as sample data. Subsequently, a basic convolutional neural network for feature point detection is trained using the synthetic graphics dataset, and further refined through homographic adaptation to enhance the generalization of feature point detection. Using this network for image annotation enables a self-supervised mechanism without the need for manual labeling.

2.1.1 Homographic Adaptation. During training, each image will be transformed by random homography transformations, the transformed images will participate in feature point detection. This process adopts an iterative training approach, continuously optimizing the detector until a large number of repetitive feature points can be detected in images with different poses after transformation. In the annotation stage, images with different poses are transformed back to their original image poses, and then added together to obtain the final annotation result. the formula is shown below:

$$x = \mathcal{H}^{-1} f_{\theta} (\mathcal{H} (I)) \quad (1)$$

$$F(I; f_{\theta}) = \frac{1}{N_h} \sum_{i=1}^{N_h} x_i \quad (2)$$

Where I represents the input image, x represents the feature point detection result, $\mathcal{H}$ represents a random homography transformation matrix, f_{θ} represents the feature point detection function, $F(I; f_{\theta})$ represents the final annotation result.

2.1.2 Loss Functions. The loss function consists of two parts, including feature point detection loss and descriptor loss. For feature point detection loss, use the cross entropy loss function to determine the distance between predicted feature points and labels:

$$L_p(X, Y) = \frac{1}{H_c W_c} \sum_{h=1}^{H_c} \sum_{w=1}^{W_c} l_p(x_{hw}; y_{hw}) \quad (3)$$

$$l_p(x_{hw}; y) = -\log \left(\frac{\exp(x_{hwy})}{\sum_{k=1}^{65} \exp(x_{hwk})} \right) \quad (4)$$

Where H_c and W_c represent the encoded feature map size, which has been pooling three times, and are one eighth of the input size. Optimizing this loss is similar to solving a multi-classification problem, where the calculation result is the probability that the sample points belong to feature points.

Descriptor loss L_d is not only used to measure the distance between the output descriptors d_{hw} and $d'_{h'w'}$ in two images before and after homography transformation, but also to determine whether the corresponding feature points are adjacent to each other:

$$L_d(D, D', S) = \frac{1}{(H_c W_c)^2} \sum_{h=1}^{H_c} \sum_{w=1}^{W_c} \sum_{h'=1}^{H_c} \sum_{w'=1}^{W_c} l_d(d_{hw}, d'_{h'w'}; s_{hwh'w'}) \quad (5)$$

$$s_{hwh'w'} = \begin{cases} 1, & \text{if } \mathcal{H} p_{hw} - p_{h'w'} \leq 8 \\ 0, & \text{if } \mathcal{H} p_{hw} - p_{h'w'} > 8 \end{cases} \quad (6)$$

Where $\mathcal{H}$ represents the homography transformation matrix between two images. As shown in the above formula, if the feature points detected in the original image after matrix transformation can match with the feature points detected in the original image after matrix transformation, the value representing the correlation is 1. The loss L_d is represented as follows:

$$l_d(d, d'; s) = \lambda_d * s * \max(0, 1 - d^T d') + (1 - s) * \max(0, d^T d' - 0.2) \quad (7)$$

2.2 Feature Points Matching With SuperGlue

SuperGlue [13] is a neural network model that can match feature points and filter outliers. The model first encodes the feature points of two images and fuses them with their respective descriptors. Through multiple iterations of attentional aggregation, it obtains more comprehensive feature description for both images. Then, based on the optimal matching, a score matrix and an allocation matrix are constructed and iteratively solved to obtain the current matching result. The optimal allocation matrix is obtained through training, so as to obtain the final feature point matching result.

2.2.1 Attentional Aggregation. Attentional aggregation refers to the fusion of feature points with their descriptors after encoding, and obtaining more comprehensive feature description through multiple alternating self-attention [14] and cross-attention. This mechanism simulates the process of humans repeatedly comparing two images, thereby capturing inner correlations and long-range

dependencies. Applying the correlation matrix output by the attention network to the feature map enables long-range information fusion.

The formula of attentional aggregation is as follows:

$$z_i = \sum_{j=1}^n \alpha_{ij} (x_j W^v) \quad (8)$$

$$\alpha_{ij} = \frac{\exp e_{ij}}{\sum_{k=1}^n \exp e_{ik}} \quad (9)$$

$$e_{ij} = (x_i W^Q) (x_j W^K)^T \quad (10)$$

Where $x = (x_1, \dots, x_n)$ represents an input sequence, and z represents an output sequence with the same size as the input sequence, α_{ij} represents the correlation coefficient of different items in the sequence, calculated by the softmax, e_{ij} used to measure the correlation between two different items.

2.2.2 Optimal Matching. The optimal matching constructs a score matrix S and an allocation matrix P according to the feature description after attentional aggregation. In the allocation matrix P , the row and column represent the feature points of the two images respectively, and the value of the matrix is the matching probability of the row and column. By iteratively solving, the current feature point matching result can be obtained.

The score matrix S describing the similarity of information can be computed by the inner product of two feature descriptor f_i^A and f_j^B :

$$S_{i,j} = \langle f_i^A, f_j^B \rangle, \forall (i, j) \in A \times B \quad (11)$$

In order to calculate the optimal allocation matrix P , the total score $\sum_{i,j} S_{i,j} P_{i,j}$ is maximized through the Sinkhorn algorithm [15].

Besides, because feature points may mismatch, It is necessary to add a row and a column to the score matrix S , which means there is no corresponding matching point. The last row and column will be removed in the score calculation, so as to filter out the external points. Finally, in a supervised learning manner, the well-trained allocation matrix P represents the final feature point matching result.

3 Algorithm

In this section, the SuperPoint feature point detection model and SuperGlue feature point matching model will be used to solve the problem of vibration feeding speed measurement. The overall pipeline of algorithm proposed in this article is shown in Figure 1.

Firstly, the video data of vibration feeding is obtained through the camera to form a dataset. Since the video data is perspective, it is necessary to correct the distortion of the data through perspective transformation, and cut out the feeding area for speed measurement. Assume that I_{dst} is the image after perspective projection transformation, and I_{src} is the source image. The formula is:

$$I_{dst} = M \cdot I_{src} \quad (12)$$

Then the network structure of SuperPoint model is constructed. The network structure is divided into two parts: the encoder and the decoder. The encoder is a VGG [16] style convolutional neural network, which contains multiple 3×3 convolution cores and $2 \times$

2 maximum pooling layers to ensure a large receptive field, and compress the input image to extract key feature information. The decoder is used to upscale the feature map through sub-pixel convolution, it outputs two feature maps. One of them is the same size as the input image, represents the probability that the point is a feature point; The other is the same as the input image in width and height, each pixel value with depth represents the feature descriptor.

After obtaining the feature points and descriptor of scrap image A and image B, the multi-layer perceptron is used to fuse the respective feature points and descriptor. Through multiple alternating self-attention and cross-attention processing, two more informative feature description f_i^A and f_j^B are obtained. Calculate the similarity score matrix S of all possible feature point matching pairs, and then use the Sinkhorn [15] algorithm to maximize the total score $\sum_{i,j} S_{i,j} P_{i,j}$ to obtain the allocation matrix P . Assuming that two images A and B have M and N feature points respectively, the loss function formula is as follows:

$$Loss = - \sum_{(i,j) \in \mathcal{M}} \log P_{i,j} - \sum_{i \in \mathcal{I}} \log P_{i,N+1} - \sum_{j \in \mathcal{J}} \log P_{M+1,j} \quad (13)$$

$\mathcal{M}$ represents the set of feature point pairs with known matching relationship, and $\mathcal{I}$ and $\mathcal{J}$ represent the set of feature points without matching in the two images. By minimizing the loss function, the final matching result can be obtained.

Finally, by matching all the feature points on the two images with the pixel displacement s_{pixel} , and combining the image height h , the actual length of the feeding area L and the unit time Δt , the real-time vibration feeding speed of scrap metal can be calculated. The formula is as follows:

$$v = -\frac{1}{N} \sum_{(i,j) \in \mathcal{M}} \frac{s_{pixel} L}{h \Delta t} \quad (14)$$

4 APPLICATION

The video dataset of scrap metal vibration feeding comes from an engineering project. When using the feature point detection model SuperPoint, we choose the pre-train model based on the COCO dataset [17]; When using the feature point matching model SuperGlue, we choose the pre-train model based on SUN3D [18] dataset which containing the camera position information. The application platform is configured as follows: the processor is Intel Xeon Silver 4210R@2.4Ghz×2, the GPU is NVIDIA RTX A4000, and the memory is 64G/DDR4.

The comparison of feature point detection between traditional Harris algorithm and SuperPoint is shown in Figure 2. It can be seen that SuperPoint has higher accuracy and robustness. Although the image is full of noise, the feature points can still be detected well, So SuperPoint is more suitable in the current complex vibration environment.

Feature point extraction and matching results before and after clipping and perspective projection transformation is shown in Figure 3(Right). It can be seen that under the same threshold setting, although more feature points can be extracted before transformation, some feature points are irrelevant to the scrap metal. After clipping and perspective projection transformation, the number of matching points increases, and the calculation of pixel displacement is easier.

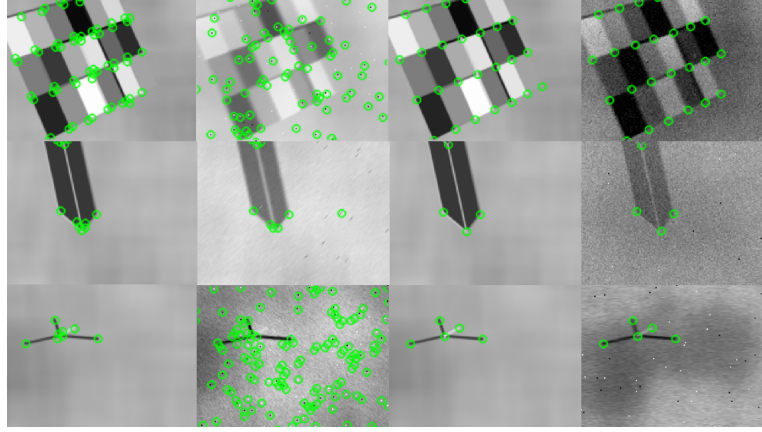


Figure 2: The comparison between the traditional Harris algorithm and SuperPoint. The first two columns: detection result of Harris on images with or without noise. The last two columns: detection result of SuperPoint on images with or without noise.

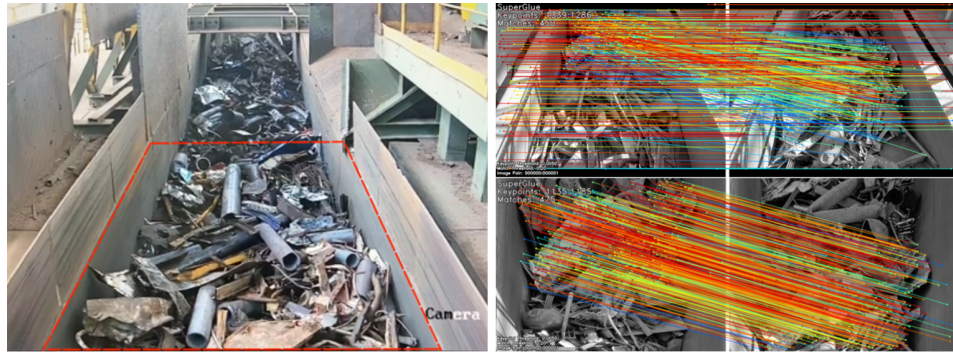


Figure 3: (Left) Scrap metal trough and feeding area, (Right) Feature point extraction and matching results before and after clipping and perspective projection transformation

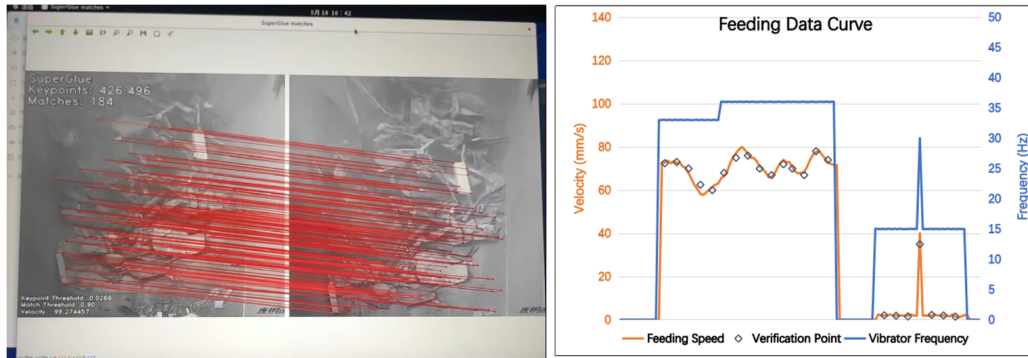


Figure 4: (Left) Application of speed measurement algorithm. (Right) Feeding data curve

The visual based speed measurement algorithm we proposed is applied in a steelworks shown in Figure 4 (Left). Figure 4(Left) displays the real-time extraction and matching of feature points between two frames with unit time interval by the speed measurement algorithm. With a high threshold setting, 426 and 496 feature points were extracted in the current scene, of which 184

pairs matched each other. Combined with the average value of the matched pixel displacement, unit time and the actual length of the feeding area, the current speed of scrap metal feeding is estimated to be 9.927 cm/s.

As shown in Figure 4 (Right), we write the estimated real-time feeding speed into the database and display it on the website. After comparison, the fluctuation of the speed curve is consistent with the frequency curve of the horizontal vibrator, and the speed are basically consistent with the verification point data measured manually.

5 Conclusions

In this article, we proposed a visual based algorithm for measuring the speed of scrap metal vibration feeding. Firstly, we collect, crop, grayscale, and transform the video stream during the feeding of scrap metal to construct a dataset. Then, by using the SuperPoint feature point detection model and the SuperGlue feature point matching model, the matched feature point pairs within a unit time are obtained. Finally, the real-time feeding speed was estimated by combining pixel displacement and the actual length of the feeding area. After on-site application, the fluctuation of the speed curve is consistent with the frequency curve of the horizontal vibrator, and the speed are basically consistent with the verification point data, effectively solved the problem in measuring the speed of scrap metal with different appearance characteristics during repeated vibration transmission.

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